Siraaj

Current Requirements and Competitor Analysis

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This revision matches the running Siraaj application and its current limits. It separates what is working in the visitor product from features that remain limited, data-dependent, or planned.

Current product direction

- The public experience is an attraction and facility discovery map for Makkah.
- The map supports English and Arabic, including English LTR and Arabic RTL.
- The current geographic map loads public reviewed records, search, categories, filters, place details, freshness, source, and external map links.
- The Haram 3D view is a location inspection companion. It does not draw indoor routes.
- Journey planning and routing code remain available for internal analysis and operator testing, but journey controls are not the main visitor map flow.
- The current seeded map reports 17 reviewed public places and facilities. Haram records remain dependent on approved evidence.

Project boundary

Siraaj standout: the Haram 3D location companion

Our 3D Haram explorer gives Siraaj a distinctive visual way to inspect gates, Mataf, Masaa, water points, toilets, medical facilities, and accessibility locations by floor. It combines spatial understanding with Arabic and English labels, source status, and verification limits.

1. Product overview and actors

Siraaj helps visitors find useful places and facilities, understand available crowd and safety information, inspect Haram locations when approved data exists, and reach external maps or manual emergency support when needed. The visitor interface uses simple decisions and honest data states instead of pretending that unavailable evidence is live or exact.

Actors

Actor	Current responsibility
User or visitor	Search attractions, view the geographic map, inspect place details, filter public records, and open an external map link.
Authenticated pilgrim	Use protected SOS, offline safety, group safety, and consent-controlled features.
Group member	View approved group information and consented safety data.
Group leader	Review group safety information and coordinate approved members without changing another member's consent.
Administrator	Maintain public records, Haram inventory evidence, safety information, incidents, and audits.

Main visitor flow

Step	Visitor experience	Current state
1	Open the attraction map or directory.	Implemented
2	Search in Arabic or English and choose a category.	Implemented
3	Select a place and inspect source, verification date, freshness, crowd label, and accessibility information.	Implemented where data exists
4	Open the location on the map or use an external map link.	Implemented
5	Use protected group, SOS, or offline safety features when authenticated.	Implemented or limited by account/data state

The 3D map is our project contribution

The map is designed as a location-understanding tool, not only a pin viewer. It gives the visitor a visual Haram context that can later grow with approved floor and facility evidence.

2. Functional requirements matched to the app

The table below is the current requirement baseline. “Implemented” means the running app has the user-facing behavior. “Limited” means the behavior exists but depends on approved data, authentication, or a controlled state. “Planned” means it is not part of the current public release.

Function	Updated requirement	Status
Attraction discovery	Browse public Holy Sites, attractions, facilities, transport points, accessibility facilities, food, essentials, and safety-related places as planning information.	Implemented
Search and categories	Search Arabic or English names, areas, and categories. Use category chips for attractions, Holy Sites, facilities, medical, water, toilets, transport, accessibility, meeting points, information desks, and emergency points.	Implemented
Place details	Show public name, Arabic/English name where available, category, area or floor, coordinates where public, crowd label, freshness, source, verification date, accessibility, and safety notices.	Implemented where data exists
Geographic map	Show reviewed public records on a MapLibre map with clustering, filters, list fallback, selected-place preview, details drawer, and external map handoff.	Implemented
Haram 3D location view	Inspect approved Haram locations by floor and category, such as gates, Mataf, Masaa, water, toilets, medical points, and accessibility facilities. Do not show indoor route lines.	Limited by approved inventory
Crowd and safety guidance	Show Comfortable, Busy, Crowded, or Not measured only when the available record supports it. Show stale, unavailable, or verify-before-visiting states honestly.	Implemented / data dependent
External map handoff	Open an external map link after a clear user action when Siraaj does not have verified location or walking coverage. Siraaj does not claim live traffic or completed navigation.	Implemented
Journey planning	Keep the routing engine for internal analysis and future work. Do not present route lines, preferences, or My Journey as the primary public attraction-map experience.	Internal / future visitor feature
Emergency support	Allow an authenticated pilgrim to create or reuse an SOS incident, show category-based emergency information, and confirm a manual call link.	Implemented
Group safety	Allow active group members to view approved group information and consented sharing. Leaders cannot silently enable or revoke another member's consent.	Implemented / protected
Offline safety	Provide an authenticated offline card, emergency numbers, last-known details, and a queued-not-delivered SOS state.	Implemented / protected
Languages and accessibility	Support English LTR and Arabic RTL, visible focus, keyboard operation, screen-reader labels, reduced motion, large text, and mobile touch targets.	Implemented and tested
Administrator maintenance	Maintain public records, sources, verification, facilities, zones, incidents, Haram inventory evidence, and audits.	Implemented / staff-only

Why the 3D map makes Siraaj different

The geographic map helps visitors find a place. The Haram 3D companion helps them understand where that place belongs inside the Holy Site. This combination supports clearer decisions while keeping approximate or unverified locations visibly limited.

3. Non-functional requirements and safety limits

Quality area	Current requirement
Security	Protected actions use authentication, CSRF protection, server-side authorization, and object-level checks.
Privacy	Public responses exclude private profiles, QR secrets, medical details, private groups, exact historical paths, and unconsented live locations.
Consent	Location sharing requires explicit consent, records policy and time, expires with the trip, and stops after revocation.
Auditability	Sensitive reads, SOS actions, consent changes, maintenance actions, exports, and resolutions are recorded where supported.
Data quality	Public records require valid coordinates, a source, verification date, enabled status, and a clear accuracy/freshness state. Unverified Haram locations remain hidden.
Honest information	Measured, simulated, stale, unavailable, and approximate values are separated. Missing evidence never becomes a fabricated route, crowd value, rating, or best time.
Availability	The map provides loading, empty, offline, unavailable, list-fallback, and stale-data states.
Performance	Public map data is compact, clustered, and usable on mobile connections. Browser storage never becomes an authorization or safety decision.
Accessibility	English LTR and Arabic RTL are supported with keyboard navigation, visible focus, screen-reader labels, reduced motion, large text, and large touch targets.
Compatibility	The public experience works on current mobile browsers and keeps a list fallback when MapLibre or WebGL is unavailable.
External handoff safety	External map and phone links open only after a clear user action. Siraaj does not claim it operated the external navigation or that an emergency call was answered.

Current implementation limits

- The attraction map is the current public map experience. Route lines and Journey controls are not shown as the primary visitor map flow.

- The Haram 3D model is location inspection only. Approved evidence is required before a Haram location is shown as verified or exact.

- The current map can show “Not measured” because crowd data is not always available.

- The current release does not claim official integrations, live CCTV, live traffic, paid map services, automatic calling, bookings, prices, ratings, or guaranteed opening hours.

A strong research and usability decision

The 3D model gives visitors a simple spatial explanation of the Haram while the geographic map provides public location details. Keeping the two views connected makes the experience clearer without inventing indoor precision.

4. Comparison with similar apps

This comparison is written from the current Siraaj position. The advantage is a focused, safety-first visitor map with clear evidence limits rather than a claim to replace every service.

App	What it provides	Siraaj focus
Nusuk	Official permits, bookings, transport, and services.	Independent attraction discovery, source-aware public records, and safety context.
Muslim Pro	Prayer, Quran, duas, Qibla, and religious content.	Movement around Makkah, facilities, accessibility, and public safety information.
Yalla Hajji	Group location, chat, geofences, SOS, and offline support.	Consent, expiry, revocation, group-only visibility, and privacy limits.
iZiyarah	Place discovery, alerts, location, offline support, and bookings.	Reviewed planning information without inventing prices, ratings, bookings, or availability.
Google Maps	Global place search, traffic, and driving navigation.	Pilgrimage-specific categories, crowd labels, Haram location evidence, and safety facilities, with external handoff when needed.

5. Why Siraaj can stand out

- Safety-first discovery: visitors see useful context before choosing a place.

- Clear evidence states: reviewed, approximate, stale, unavailable, and Not measured are visible.

- Haram location focus: the 3D model can become a trustworthy location companion after approved inventory is supplied.

- Consent-based group safety: coordination does not silently become tracking.

- Simple emergency support: the app records the request and opens a manual call link only after confirmation.

- Honest scope: Siraaj does not imitate official services or claim live capabilities it cannot verify.

6. Sources

Nusuk services: https://www.nusuk.sa/all-services
Muslim Pro features: https://www.muslimpro.com/features/
Hajj and Umrah App: https://www.hajjumrahapp.com/en
Yalla Hajji: https://www.yallahajji.com/mobile-app
iZiyarah: https://iziyarah.com/

Revision note

This document updates the earlier planned requirements to match the current visitor map, English/Arabic implementation, Haram 3D location direction, privacy boundaries, and current feature status. Future route work remains separate from the verified location inventory.